ستاجير
كُلّ مَا تحتاجه في الطب

ENDOCRINOLOGY

Parathyroid Gland

Internal Medicine Lecture Series

- Parathyroid Hormone (PTH) play a key role in the regulation of calcium and phosphate homeostasis and vitamin D metabolism.

Classification of diseases of parathyroid glands:

1. Hormone excess.
2. Hormone deficiency.
3. Hormone hypersensitivity.
4. Hormone resistance.
5. Non functioning tumours .

20.30 Classification of diseases of the parathyroid glands

	Primary	Secondary
Hormone excess	Primary hyperparathyroidism Parathyroid adenoma Parathyroid carcinoma[1] Parathyroid hyperplasia[2] Tertiary hyperparathyroidism Following prolonged secondary hyperparathyroidism	Secondary hyperparathyroidism Chronic kidney disease Malabsorption Vitamin D deficiency
Hormone deficiency	Hypoparathyroidism Post-surgical Autoimmune Inherited	
Hormone hypersensitivity	Autosomal dominant hypercalciuric hypocalcaemic (CASR-activating mutation)	
Hormone resistance	Pseudohypoparathyroidism Familial hypocalciuric hypercalcaemia	
Non-functioning tumours	Parathyroid carcinoma[1]	

[1]Parathyroid carcinomas may or may not produce parathyroid hormone. [2]In multiple endocrine neoplasia (MEN) syndromes (p. 700).
(CASR = calcium-sensing receptor)

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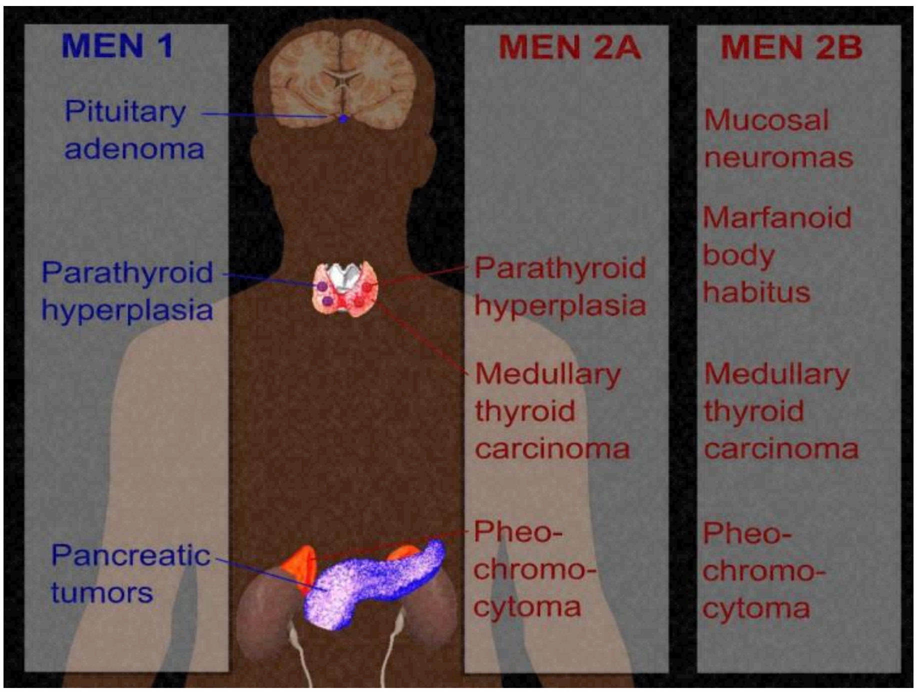
- Caused by autonomus secretion of parathyroid hormone usually by single parathyroid adenoma, which can vary in diameter from a few millimeters to several cm.
- It is more common in females.
- May be part of MEN Syndrome .
- (multiple endocrine neoplasia) , in which case multiple nodules are present .



Clinical presentation

- Symptoms are those of hypercalcemia , bone pain & tenderness is common , fractures & even deformity.
- Ostietis fibrosa cystica : result from increased bone resorption by osteocalsts with fibrous replacement in the lacunae.
- chondrocalcinosis : may occu due to deposition of calcium pyrophosphate crystals within rticular cartilage.
- Recurrent stone formations .
- Nephrocalcinosis : less common than other presentations.

1. **Biochemical** : There is high PTH , high s. calcium with low or normal s. phosphate .
2. **Radiological**: Skeletal X ray may be normal in mild cases , but in patient with advanced disease there may be subperiosteal erosions & terminal resorption in the phalanges .

   Pepper pot spots on skull X ray In nephrocacinosis , scattered opacities may be visible on plain abdominal x ray.

   DXA scan is to detect osteoporosis.

   Localization of the adenoma is done by parathyroid sestamibi scan specially before surgery .

   Other imaging: ultrasound of the neck , MRI ..

i 20.35 Hyperparathyroidism

Type	Serum calcium	PTH
Primary Single adenoma (90%) Multiple adenomas (4%) Nodular hyperplasia (5%) Carcinoma (1%)	Raised	Not suppressed
Secondary Chronic renal failure Malabsorption Osteomalacia and rickets	Low	Raised
Tertiary	Raised	Not suppressed

Management

- The treatment of choice for primary hyperparathyroidism is surgery.
- **First** : correct the hypercalcemia (medical emergency) (will be discussed later in this lecture .)

Indications of surgery :

1. Age less than 50 with clear symptoms .
2. Presence of complications like renal stones , renal impairment & osteoporosis .
3. Asymptomatic patients with significant hypercalcemia (s. calcium more than 11.4 mg /dl .

- Be ware of significant hypocalcemia postoperatively specially those with bone disease .
- Cinacalcet can be used in patients not fit for surgery.

The most common cause of hypoparathyroidism is damage to the gland during thyroid surgery , rarely can occur as a result of infiltration of the gland by copper or iron (wilson , hemochromatosis) or congenital causes.

- **Clinical features** : features of hypocalcemia (discussed in the next slides) .
- **Lab. Findings :** low PTH , low calcium , high phosphate .
- **Management** : usually treated with oral calcium & vitamin D analogues like 1 alpha hydroxycholecalciferol (alfacalcidol) or calcitriol.

Hypercalcaemia

- It is common biochemical finding that can be asymptomatic , or my be presented as acute emergency cases or with chronic symptoms .
- By far the most common causes are primary hyperparathyroidism & malignant disease .
- Other causes shown in the table (next slide) .

i 20.32 Causes of hypercalcaemia
With normal or elevated parathyroid hormone (PTH) levels
• Primary or tertiary hyperparathyroidism • Lithium-induced hyperparathyroidism • Familial hypocalciuric hypercalcaemia
With low PTH levels
• Malignancy (lung, breast, myeloma, renal, lymphoma, thyroid) • Elevated 1,25-dihydroxyvitamin D (vitamin D intoxication, sarcoidosis, human immunodeficiency virus, other granulomatous disease) • Thyrotoxicosis • Paget's disease with immobilisation • Milk-alkali syndrome • Thiazide diuretics • Glucocorticoid deficiency

- Common symptoms include : polyuria , polydipsia , renal colic , lethargy , nausea , peptic ulcer , constipation , depression & impaired cognition.
- Malignant hypercalcemia usually has rapid onset
- Remember the classic (bones , stones , abdominal groans) ! Although not present in all patients , because 50 % my be asymptomatic & discovered accidentally by routine blood tests
- Family history of hypercalcemia may suggest **MEN Or familial hypocalciuric hypercalcemia.!**

Painful Bones	Painful bone condition (Classically osteitis fibrosa cystica)
Renal Stones	Kidney Stones (Can ultimately lead to Renal failure)
Abdominal Groans	GI symptoms: Nausea, Vomiting, Constipation, Indigestion
Psychiatric Moans	Effects on nervous system: lethargy, fatigue, memory loss, psychosis, depression

Management of severe hypercalcaemia

1. **Intravenous 0.9 % saline** : this is the first important step to correct dehydration , improve renal function & increase urinary calcium excretion .
2. **Bisphosphonates** : zolidronic acid 4 mg iv infusion or pamidronate 60-90 mg iv , to inhibit bone resorption & it can reduce s. calcium within 3-5 days .
3. **Calcitonin :** acts rapidly to increase calcium excretion & reduce bone resorption . & can be combined with bisphosph. & iv fluid for the 1 st 24 -48 hrs. in patient with life threatening conditions.
4. **Denosumab :** in severe cases , refractory to other treatment .

Aetiology :

- The most common cause of hypocalcemia is low serum albumin with normal ionised calcium concentration .
- The other cause is alkalosis (here total calcium may be normal but with low ionised calcium)
- Low magnesium (should be suspected in patient with malabsorption , diuretic therapy & alcohol excess, it lead to hpocalcemia by impairing the Parathyroid g. to secrete PTH.
- others: vit. D deficiency , Chronic renal failure , hypoparathyroidism ..

i 20.33 Differential diagnosis of hypocalcaemia

	Total serum calcium	Ionised serum calcium	Serum phosphate	Serum PTH	Comments
Hypoalbuminaemia	↓	↔	↔	↔	Adjust calcium upwards by 0.02 mmol/L (0.1 mg/dL) for every 1 g/L reduction in albumin below 40 g/L
Alkalosis	↔	↓	↔	↔ or ↑	p. 632
Vitamin D deficiency	↓	↓	↓	↑	p. 1053
Chronic renal failure	↓	↓	↑	↑	Due to impaired vitamin D hydroxylation Serum creatinine ↑
Hypoparathyroidism	↓	↓	↑	↓	See text
Pseudohypoparathyroidism	↓	↓	↑	↑	Characteristic phenotype (see text)
Acute pancreatitis	↓	↓	↔ or ↓	↑	Usually clinically obvious Serum amylase ↑
Hypomagnesaemia	↓	↓	Variable	↓ or ↔	Treatment of hypomagnesaemia may correct hypocalcaemia

(↑ = levels increased; ↓ = levels reduced; ↔ = levels normal)

Clinical presentation

- Mild hypocalcaemia is often asymptomatic , but with more profound reduction of calcium , Tetany occur (characterised by muscle spasm due to increased excitability of peripheral nerve
- Children are more liable to develop tetany & may present with characteristic triad of (carpopedal spasm , convulsion & stridor) .
- The adults present with carpopedal spasm, tingling of hand, feet & around the mouth .

sphygmomanometer cuff on the upper arm will result in carpopedal spasm within 3 minutes.)

other sign is chvostic sign , in which tapping over the branches of facial n.

As they merge from the parotid g result in twitching of the facial muscles.

- Hypocalcaemia in severe cases can cause papillodema , prolongation of QT interval on ECG leading to arrhythmia .
- Prolonged hypocalcemia & hyperphosphatemia (as in hypoparathyroidism) can lead to calcification of basal ganglia , epilepsy , psychosis & cataract .
- Hypocalcemia with hypophosphatemia (as in vit D deficiency) can cause osteomalacia in adults & rickets in children.

20.34 Management of severe hypocalcaemia

Immediate management

- 10–20 mL 10% calcium gluconate IV over 10–20 mins
- Continuous IV infusion may be required for several hours (equivalent of 10 mL 10% calcium gluconate/hr)
- Cardiac monitoring is recommended

If associated with hypomagnesaemia

- 50 mmol (1.23 g) magnesium chloride IV over 24 hrs
- Most parenteral magnesium will be excreted in the urine, so further doses may be required to replenish body stores

(IV = intravenous)


SPOT DIAGNOSIS
@Medcounter
A Tetany
B Chvostek's sign
C Trousseau's sign
D Lhermitte sign
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What is the name of the sign shown below?
@Medcounter
A Hypocalcemia
B Chvostek's sign
C Trousseau's sign
D Lhermitte sign
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